

1 Worksheet

Question 1.1. Let $f : E \rightarrow [0, \infty]$ satisfy $\int_E f < \infty$. Let $A_n \subseteq A_{n-1} \subseteq E$ be a decreasing sequence of sets such that $m(A_n) \rightarrow 0$ as $n \rightarrow \infty$. Show that

$$\lim_{n \rightarrow \infty} \int_E f \mathbb{1}_{A_n} = 0$$

Proof.

□

Question 1.2. Let E be Lebesgue measurable with $m(E) > 0$. Show that there exists $p \in E$ so that $m(E \cap \{x : |x - p| < \varepsilon\}) > 0$ for all $\varepsilon > 0$. (Problem suggested by a student.)

Proof.

□

Question 1.3. Let $f_n : \mathbb{R} \rightarrow [0, \infty]$ be a sequence of Lebesgue measurable functions that converge pointwise to some f . Assume that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f < \infty$$

Show that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} \mathbb{1}_E f_n = \int_{\mathbb{R}} \mathbb{1}_E f$$

for every measurable E . Show that this may fail if $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f = \infty$.

Proof.

□

Question 1.4. Let $f_n : E \rightarrow [0, \infty]$ be a decreasing sequence of Lebesgue measurable functions such that $\int_E f_1 < \infty$. Let f be the pointwise limit. Show that

$$\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$$

Proof.

□

Question 1.5. If $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is integrable, show that the function

$$F(x) := \int_{\mathbb{R}} \mathbb{1}_{(-\infty, x)} f$$

is continuous.

Proof.

□